D-01: Stored plate-reader fluorescence signals

What numerical and spatial patterns are present in the supplied native results?

Materials and methods

Three original VMR result files, one VWP protocol and source-specific cadmium experiment notes.

Decompress JSON and map values with the stored physical row/column indices. Summarize each GUID/gain separately.

Plot independent native-value colour scales. No gain conversion, condition assignment, blank or OD correction.

Scope and working assumptions

Each VMR GUID and gain defines a separate stored result set. Explicit file coordinates define the wells, and values retain their native scale. The archive has no accepted well-to-clone/dose/control map or gain/flag calibration; source records also differ in instrument naming. This analysis reports native signals without blank, biomass or dose-response correction.

Results

Recovered 150 readings across 3 distinct stored results, with 50 physical wells A1:E10 each.

gain 2939: range 23036-260000, 29/50 entries equal 260000; gain 2976: range 27641-260000, 13/50 entries equal 260000; gain 933: range 22-3328, 0/50 entries equal 260000.

Gain	Wells	Median	260000 count	Input data	Finished without more wet-lab data?
2976	50	195949	13	Original VMR files	Yes: raw description
2939	50	260000	29	Original VMR files	Yes: raw description
933	50	110	0	Original VMR files	Yes: raw description

Interpretation

The three stored result sets contain 150 readings across wells A1-E10. Native signals vary across wells and gain settings. The exported readings and coordinates provide a reproducible numerical record of the plate measurements.

References and reproducibility

Original VMR ProtocolResult fields and cadmium_experiment_notes.md: R-WET-12AUG-VOYAGER / R-WET-12AUG-LAYOUT.

Complete inline code: tasks/D-01/D-01.ipynb. Numerical data and source/software provenance: data/output/D-01/.

Figures

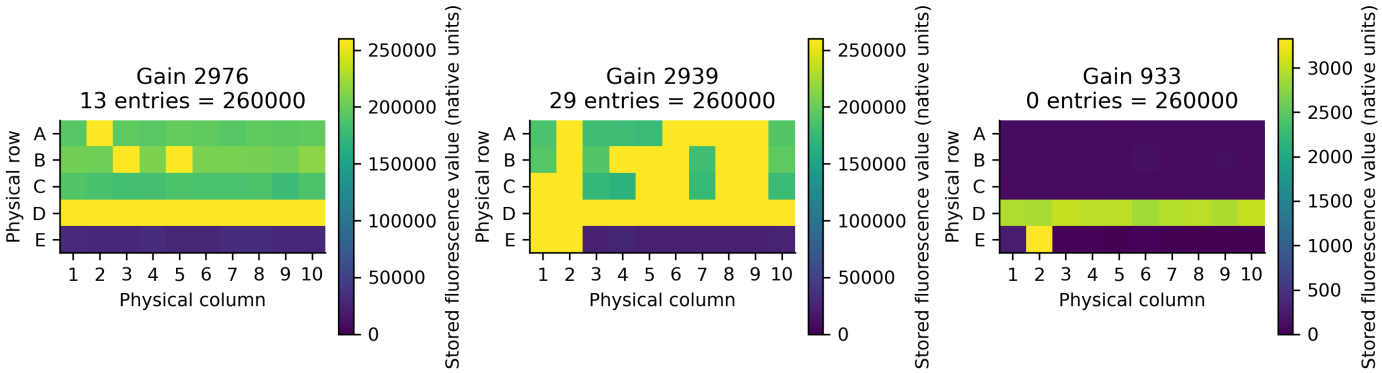


Figure 1. Stored fluorescence by physical well. Each result uses its own colour scale; gains differ. No condition labels or uncertainty estimates are assigned.